Module I INTRODUCTION

Electric energy is necessary for industrial, commercial, domestic, agricultural, scientific, transport, social, defence activities etc. It is very important to note that day to day life come to stand-still without electricity. Energy needs are supplied in the form of petrol, diesel, fire wood and electricity. Electrical energy is preferred normally due to its high efficiency and cleanliness.

Power System is a combination of Generation, Transmission and Distribution System. Electricity is generated at Electrical Power Generating Stations or Power Plants, Transmitted thorough Transmission Systems and then it is distributed to consumers from Distribution Systems. The Structure of Power System using Single Line Diagram (**SLD** and also called as One Line Diagram **OLD**) is given below:

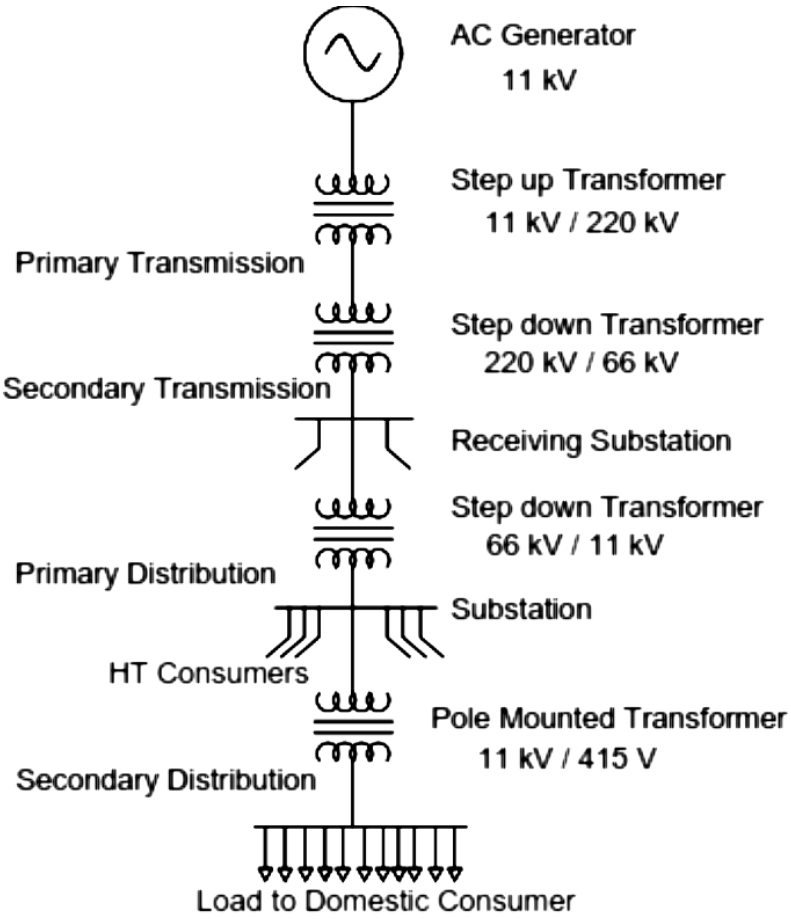


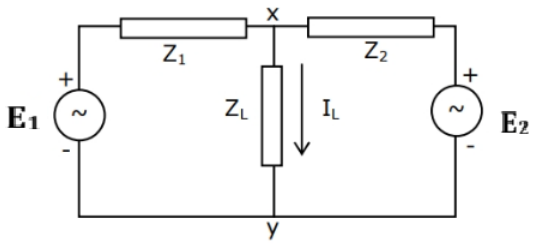
FIG : Single Line Diagram of Power Supply System

Electric Network or Electric Circuit: It is interconnected arrangement of elements like Generator or Voltage Source, Load, Resistors, Inductors, Capacitors, Impedances etc. If the circuit is closed, an electric current flows through it.

Types of Electric Circuits : Two types of currents may flow in an electric circuit i) Direct current ii) alternating current. And accordingly we have D.C circuits and A.C circuits.

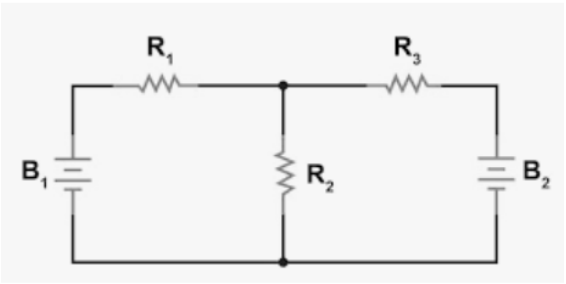
A D.C circuit consists of constant DC voltage sources, constant current sources and their interconnection with resistors, inductors and capacitors. Similarly, A.C circuit consists of Constant Voltage Sources or AC Generators, Constant Current Sources and their

interconnection with resistances, inductors and capacitors, impedances and admittances. A simple DC and AC circuit is shown below.



X, Y are Nodes or junctions
Z_1, Z_2, Z_L are impedances
E1, E2 are AC Voltage Source or AC Generators

Fig. AC Circuit or AC Network



R_1, R_2, R_3 are resistors
B_1, B_2 are Batteries

Fig. DC Circuit or DC Network

Electric Current : It is defined as rate of flow charge or electrons through a conductor.

$$i = \frac{dq}{dt} \qquad \text{A}$$

Where

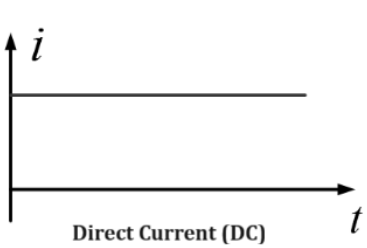
q = Charges of electrons flowing through a conductor. Charge of an electron = 1.602×10^{-19} C

t = Time interval of flow of electrons in seconds.

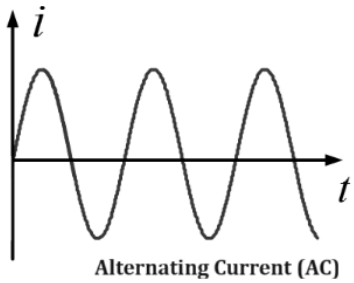
The unit of current is Ampere (A).

Types of Electric Current : The two main types of electric current are Direct Current (DC) and Alternating Current (AC).

a) DC has a constant magnitude and flows in a single direction, like that from a battery. It used in low-voltage applications such as electronic devices like smart phones, calculators etc. vehicles



Direct Current (DC)



Alternating Current (AC)

b) AC, on the other hand, has a time-varying magnitude and periodically reverses direction. It is used in homes, industries etc.

Voltage : It is also called as measure of potential difference or electric pressure that pushes electric charges through a conductor in an electric circuit.

Voltage is defined as the difference in electric potential between two points in an electrical circuit. It represents the amount of energy (W) required to move a unit of electric charge (q) between two points. Its unit is Volts (Volts or V).

$$\text{Voltage} = \frac{W}{q} \quad \text{Volts}$$

Electrical Power (P): It is defined as rate at which electrical energy is transferred or consumed in an electrical circuit or device. Its unit is watts (W).

$$Power\ P = \frac{Energy}{Time} = \frac{\Delta W}{\Delta t} \qquad Watts$$

Note : In a water pipe, the voltage is like the water pressure, and the current is like the amount of water flowing through the pipe per second. Power, then, is a combination of both the pressure (voltage) and the flow rate (current).

Electric power (P) = V * I

Where

Power is measured in watts (W), Voltage in volts (V), and Current in amperes (A)

Electrical Energy (W) : Energy is the capacity or ability to do work. Electrical energy is defined as energy generated by the movement of electrons from one point to another in an electric circuit. Its unit is **watt-sec** or **joules.**

$$\boldsymbol{Energy \quad W = Power \times time = P \times t \qquad watt - sec \ \ or \ Joules}$$

E = V×I×t Wsec (Watt Seconds)

One kWh is called as One Unit of Energy

$$\text{One Unit of Energy} = \frac{P \times t}{1000 \times 3600} \quad kWh$$

$$\boldsymbol{Energy\ E = \frac{V\ I\ t}{1000 \times 3600} \qquad kWh}$$

Ohm's Law: It states that, potential difference between two ends of conductor (V) is directly proportional current (I) flowing through it, provided the temperature remains constant.

$$\mathbf{V\ \alpha\ I}$$

$$\mathbf{V = I\ R} \qquad \mathbf{\ldots\ldots\ldots\ldots\ldots(1)}$$

Where

R is a constant known as the Resistance of the conductor.

Resistance (R) is the property of the material which resists the flow of charges through it. It is denoted by (R) and is measured in ohms (Ω).

Limitations of Ohm's law:

1. Ohm's law doesn't hold good for non-metallic conductors such as silicon carbide.
2. Ohm's law doesn't hold good for non-linear devices such as zener diodes, voltage regulators etc.

Note : **Another way of defining Ohms Law:** It states that, current flowing through the conductor is directly proportional to potential difference between two ends of conductor, provided the temperature remains constant.

$$\mathbf{I \ \alpha \ V}$$

$$\mathbf{I = G\,V} \quad \text{or} \quad V = \frac{I}{G} \qquad \text{.....(2)}$$

Where

G is a constant known as the **Conductance of the conductor.**

$$\mathbf{G = \frac{I}{V}} \qquad \textbf{Siemens or mhos}$$

Conductance : It is the property of the material to allow flow of current through it. Its unit is mhos or Siemens.

From the equations (1) and (2), $G = \frac{1}{R}$

Resistor : It is defined as a two-terminal passive element that provides electrical resistance to current flow.

- In electrical circuits, the Resistor will be represented as



- Resistance is the property of Resistor.
- Resistance is denoted by R.
- The unit of Resistance is Ohm (Ω).
- Ohm was named as the unit of Resistance in the honour of Georg Simon Ohm, a Physicist and Mathematician from Germany.

For any conductor, $R = \frac{\rho L}{A}$

Here, R = Resistance of the Conductor

ρ = Resistivity or Specific Resistance of the Conductor

L = Length of the Conductor

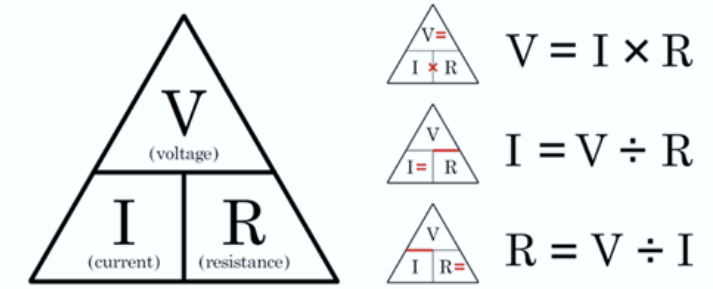
A = Area of Cross-Section of the Conductor

The Power absorbed by the Resistor will be dissipated in the form of Heat.

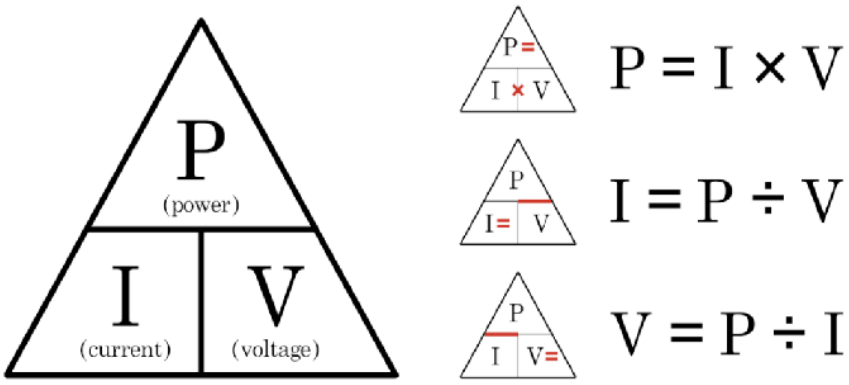
If L = 1 m, A = 1 m^2, R = ρ.

Specific Resistance or Resistivity is defined as resistance between the opposite faces of a metre cube of that material.

Ohms Law Traingle



Power Formula Traingle



Problem: Determine the specific resistance of a metal wire of 2m length and 0.6mm diameter, if the resistance of the wire is 50Ω.

Solution:

Given Data,

Length = L = 2 m

Diameter = d = 0.6 mm = $(0.6 * 10^{-3})$ m

Resistance = R = 50 Ω

Radius $r = \frac{d}{2} = \frac{(0.6 * 10^{-3})}{2} = (0.3 * 10^{-3})$ m

Area of Cross-Section $A = \pi r^2 = 3.14 * (0.3 * 10^{-3})^2 = (0.28 * 10^{-6})$ m^2

Resistance $R = \frac{\rho L}{A}$

$\Rightarrow \quad 50 = \frac{\rho * 2}{(0.28 * 10^{-6})}$

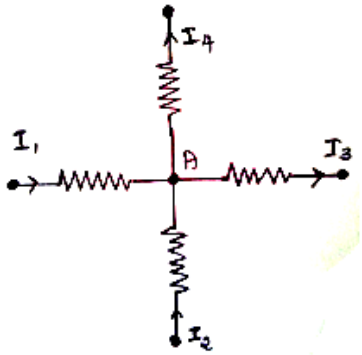
$\Rightarrow \quad \rho = (7 * 10^{6})$

∴ Specific Resistance $\rho = (7 * 10^{6})$ Ω - m

Kirchoff's Laws

Kirchoff's Current law (KCL): It states that, in an electrical circuit the algebraic sum of currents at a junction is zero

$$\text{i.e. } \sum I = 0$$



Consider the circuit shown below. The junction 'A' of the circuit has four currents I_1, I_2, I_3, I_4.

All the currents flowing towards the junction are taken as positive (assigned with positive sign).

All the currents flowing away from the junction are taken as negative (assigned with positive sign).

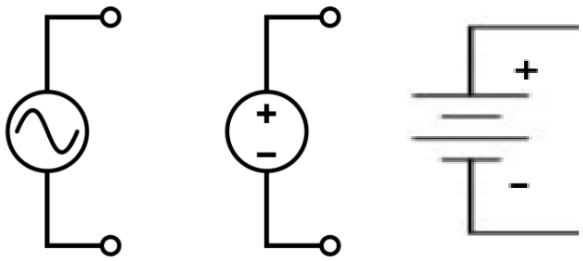
Then, according to KCL,

$$\mathbf{I_1 + I_2 - I_3 - I_4 = 0}$$

or

$$\mathbf{I_1 + I_2 = I_3 + I_4}$$

Electromotive Force (EMF or emf) : It is the energy supplied by a source (like a battery or generator) to push electric charges through a circuit. It is represented by letter 'E' or 'e', and expressed in volts.



Kirchoff's Voltage Law (KVL): It states that, in an electrical circuit, the algebraic sum of all the emf's and the voltage drops is equal to zero

$$\text{i.e. } \sum E + \sum IR = 0$$

Note : All the voltage rises are taken as positive and all the voltage drops are taken as negative.

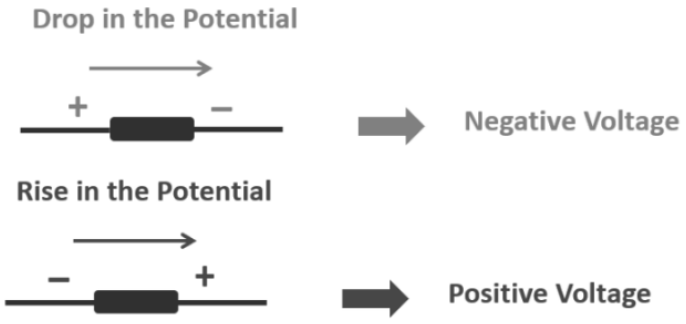
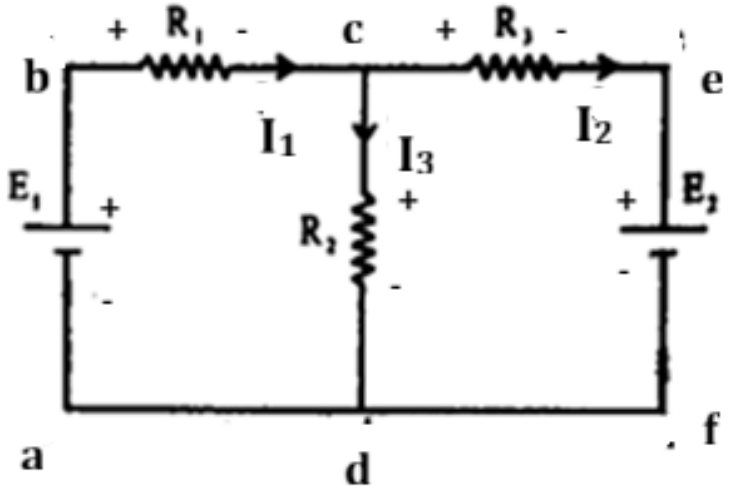


Fig. Sign Conventions for the Kirchhoff's Voltage Law

Consider a circuit shown below



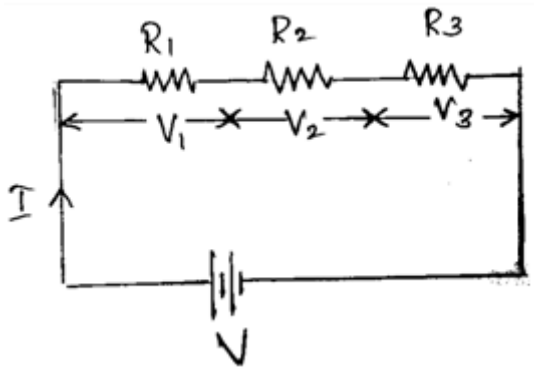
The directions of the currents I_1, I_2, I_3 flowing in the various branches are assumed.

For the closed loop in this circuit, using KVL the equations are:

For loop abcda, $E_1 - I_1R_1 - I_2R_2 = 0$

For loop dcefd, $I_2R_2 - I_3R_3 - E_2 = 0$

Analysis of Series Circuit : Consider the circuit shown below,



here, three resistances R_1, R_2 and R_3 are connected in series with DC voltage source.

V is the total voltage applied across the series combination

I is the current flowing through the circuit

V_1, V_2 and V_3 are the voltage drops cross R_1, R_2 and R_3 respectively.

As per KVL,

Then,

$$V - V_1 - V_2 - V_3 = 0$$

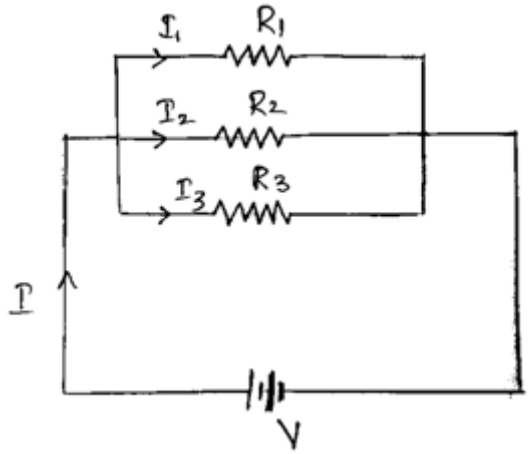
$$V = V_1 + V_2 + V_3$$

$$I R_{eq} = I R_1 + I R_2 + I R_3$$

or $$\mathbf{R_{eq} = R_1 + R_2 + R_3}$$

∴ Total Resistance in an electrical network when the resistances are connected in series is sum of all the resistances

Analysis of Parallel Circuit: Consider the circuit shown below,



Here, three resistances R_1, R_2 and R_3 are connected in parallel across a voltage V.

I is the total current

I_1, I_2 and I_3 are the currents flowing through R_1, R_2 and R_3 respectively.

$$\mathbf{I = I_1 + I_2 + I_3}$$

Or $$\frac{V}{R_{eq}} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

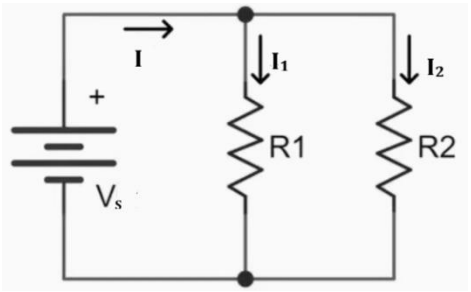
Where R_{eq} = Total resistance and is given by

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

For two resistance in parallel $R_{eq} = \frac{R_1.R_2}{R_1+R_2}\ \Omega$

Current Division Rule :

A parallel circuit acts as a **current divider** as it divides the total circuit current in its all branches as shown below.



According to Ohm's law,

$$I_1 = \frac{V}{R_1} \text{ and } I_2 = \frac{V}{R_2}$$

Let R is the equivalent resistance of the circuit, and it is given by,

$$R = \frac{R_1 R_2}{R_1 + R_2} \quad \cdots (1)$$

Also, from the circuit, we get,

$$I = \frac{V}{R} = V \times \left(\frac{R_1 + R_2}{R_1 R_2}\right) \quad \cdots (2)$$

But, we know that the voltage across both resistances is same.

$$\therefore V = I_1 R_1 = I_2 R_2 \quad \cdots (3)$$

Hence, from equations (2) & (3), we finally get,

$$I = I_1 R_1 \left(\frac{R_1 + R_2}{R_1 R_2} \right) = I_1 \left(\frac{R_1 + R_2}{R_2} \right)$$

$$\therefore I_1 = \frac{IR_2}{R_1 + R_2} \quad \cdots (4)$$

Similarly,

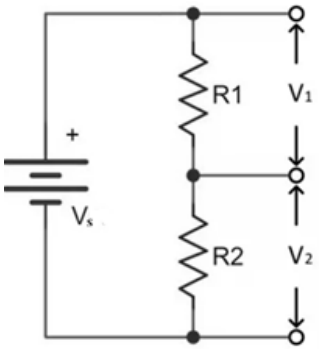
$$I = I_2 R_2 \left(\frac{R_1 + R_2}{R_1 R_2} \right) = I_2 \left(\frac{R_1 + R_2}{R_1} \right)$$

$$\therefore I_2 = \frac{IR_1}{R_1 + R_2} \cdots (5)$$

From the above equations it is seen that, ***as per current division rule, the current in any of the parallel branches is equal to the ratio of opposite branch resistance to the sum of all resistances, multiplied by the total circuit current.***

From (4) and (5) equations, **$I = I_1 + I_2$**

<u>Voltage Division Rule</u> : A series circuit acts as a voltage divider as it divides the total supply voltage into different voltages across the circuit elements as shown below.



According to Ohm's law,

$$V_1 = IR_1 \text{ and } V_2 = IR_2$$

Let R is the total resistance of the circuit, and it is given by,

$$R = R_1 + R_2 \quad \cdots (1)$$

Also, from the circuit, we have,

$$V = IR = I(R_1 + R_2) \quad \cdots (2)$$

But

$$I = \frac{V_1}{R_1} = \frac{V_2}{R_2} \quad \cdots (3)$$

Therefore, from equations (2) & (3), we finally get,

$$V = \frac{V_1}{R_1}(R_1 + R_2)$$

$$\therefore V_1 = \frac{VR_1}{R_1 + R_2} \quad \cdots (4)$$

Similarly,

$$V = \frac{V_2}{R_2}(R_1 + R_2)$$

$$\therefore V_2 = \frac{VR_2}{R_1 + R_2} \quad \cdots (5)$$

Hence, from equations of ***voltage division rule, it may be said that the voltage across a resistor in a series circuit is equal to the product of the value of that resistor and the total supply voltage, divided by the total resistance of the series resistors***.

Electrical Source : It is a device which converts mechanical, chemical, thermal or some other form of energy into Electrical Energy. In other words, the source is an active network element used for **Generating Electrical Energy or Electricity**.

Electricity : It is a fundamental form of energy characterized by the presence and flow of electrons.

Voltage Source : It is a two-terminal device whose voltage at any instant of time is constant and is independent of the current drawn from it. Such a voltage source is called an **Ideal Voltage Source** and has zero internal resistance as shown below. But, practically an ideal voltage source cannot be obtained.

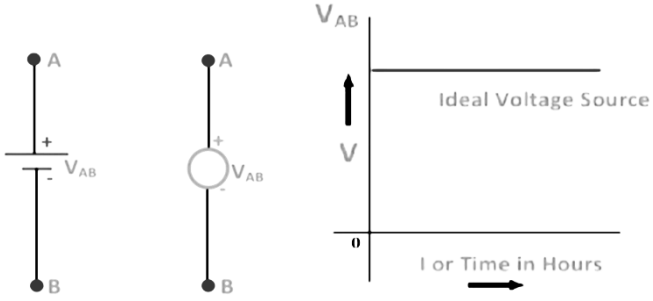


Fig : Ideal Voltage Source and its characteristics

In practice, voltage sources are having some amount of internal resistances ('r' ohms), such voltage sources are known as **Practical Voltage Source**. Due to this internal resistance; voltage drop takes place, and it causes the terminal voltage to reduce as shown below.

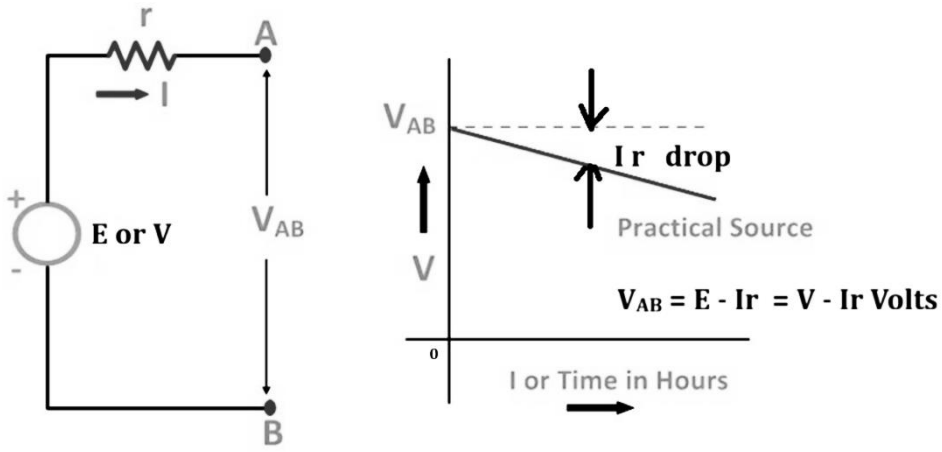


Fig : Practical Voltage Source and its Characteristics

Ex : Batteries, DC Generators, Alternators or AC Generators are practical voltage sources

Current Source : These sources are classified into Ideal and Practical current sources.

Ideal Current Source : It is a two-terminal circuit element which supplies the same or constant current, independent of its output voltage and It has infinite resistance as shown below.

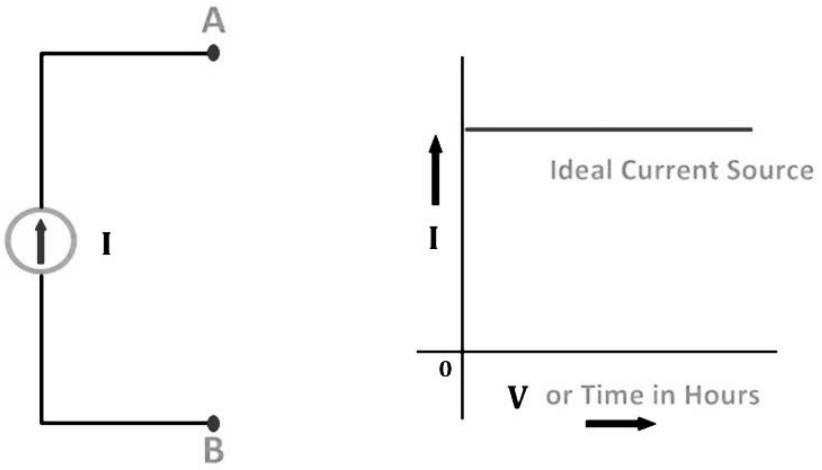


Fig : Ideal Current Source and its Characteristics

Practical Current Source : It is modelled or represented as an ideal current source in parallel with a internal resistance as shown below.

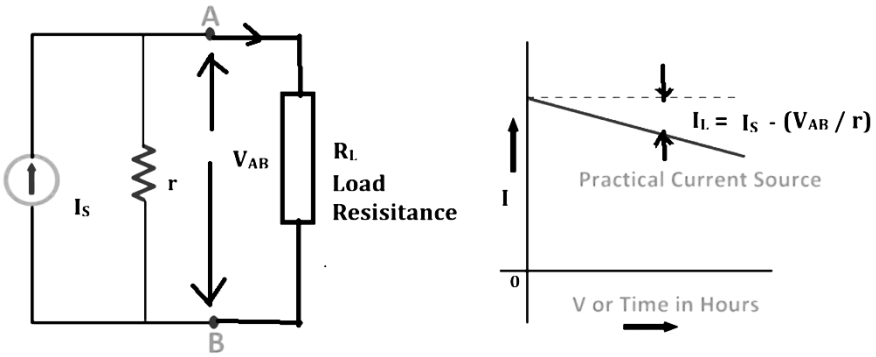
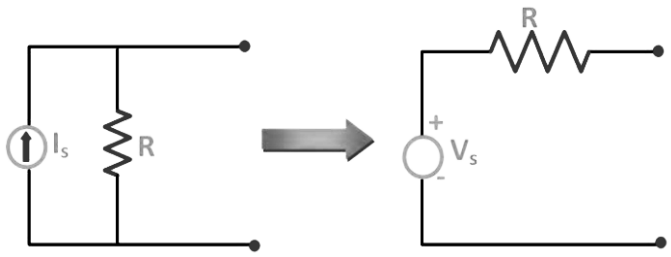


Fig : Practical Current Source and its Characteristics

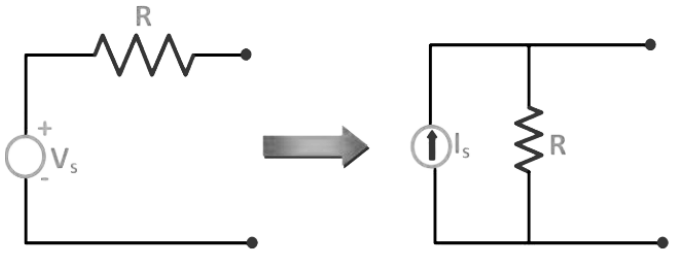
Source Transformation: It is a method of replacing or conversion of one source by an equivalent source. It means conversion of Voltage Source by its equivalent Current Source or vice versa.

Conversion of Current Source into Voltage Source: In this method of transformation a current source (I_S) which is connected in parallel with the resistance (R) is transformed into a voltage source (V_s) by placing the resistance in series with the voltage source as shown below:



Where, $\boldsymbol{V_S = I_S R}$ $\boldsymbol{Volts}$

Conversion of Voltage Source into Current Source: In this method of transformation, the voltage source (V_S) connected in series with the resistance (R) is converted to its equivalent current source (I_S) parallel with the resistance (R) as shown below.



Where, $\boldsymbol{I_S = \frac{V_S}{R}}$ $\boldsymbol{Amps}$

Electromagnetic Induction: It is the process of producing or inducing an EMF and hence an electric current when there exist relative motion between magnetic field and conductor as shown below.

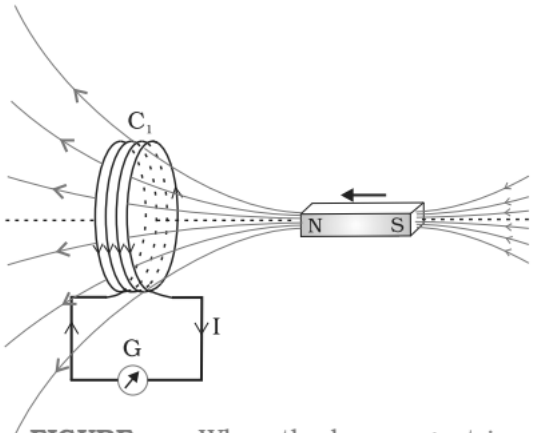
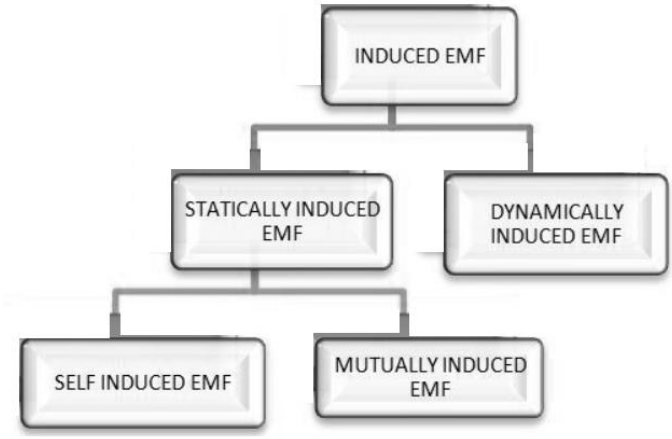


FIGURE : When the bar magnet is pushed towards the coil, the pointer in the galvanometer G deflects.

Various types of EMF induced due to relative motion between coil and magnetic field are shown below.



Faradays Laws of Electromagnetic Induction : The phenomenon of electromagnetic induction is explained in following two laws :

First Law : It states that, whenever a magnetic flux linking with a coil changes, an EMF is induced in it. or Whenever a conductor cuts magnetic field, an EMF is induced in it.

Second Law : It states that, the magnitude of induced EMF (e.m.f) is equal to rate of change of flux linkages with the conductor.

Magnitude of induced e.m.f $\boldsymbol{e = N\frac{d\emptyset}{dt}}$ **Volts**

Where

N = Number of turns of the coil

Φ = Magnetic Flux = B.A webers or wb

B = Magnetic field strength wb/m^2

A = Area of the coil m^2

dt = Time interval in seconds = $t_2 - t_1$ seconds or 's'

dϕ = Change of magnetic flux or flux = $\phi_2 - \phi_1$

$\boldsymbol{\frac{d\emptyset}{dt} = Rate\ of\ Change\ of\ flux\ with\ the\ coil\ of\ N\ turns}$

Lenz's Law : It states that, the induced EMF due to a change of flux linkage by a coil will produce a current in the coil in such a direction that it will produce a magnetic field which will oppose the cause, that is change of flux linkages.

$$e = -N\frac{d\phi}{dt} \qquad \text{Volts}$$

The minus sign indicates that the induced emf sets up a current in such a direction that magnetic effect produced by it opposes the very cause produced it.

Right Hand Grip Rule : If we hold the current carrying conductor in our right hand, then the thumb shows the direction of current while the fingers encircling the conductor shows the direction of magnetic field lines of force.

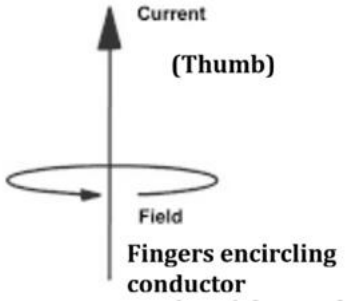
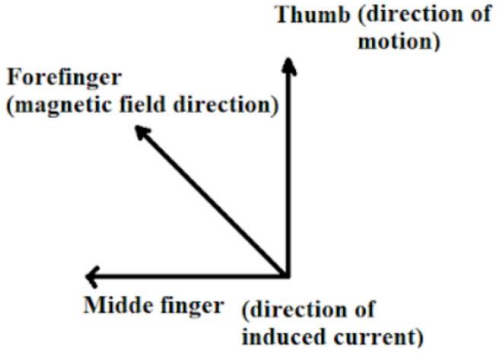


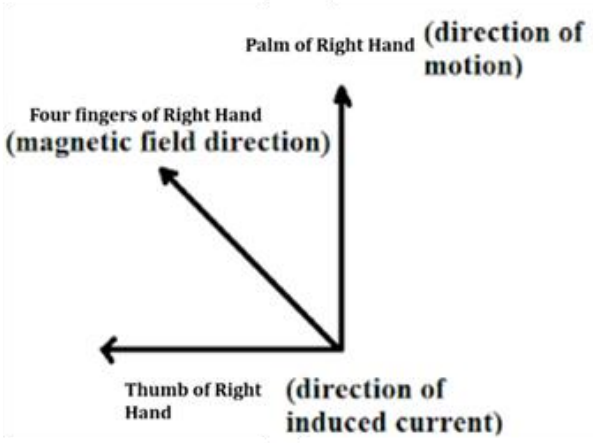
Fig : Right Hand Grip Rule

Flemings Right Hand Rule: It helps to find the direction of an induced current in a conductor moving in a magnetic field. As per this rule, if we stretch the thumb, forefinger and middle finger of our right hand such that they are mutually perpendicular to each other, then forefinger point indicates the direction of the magnetic field and the thumb indicates the direction of motion of the conductor, then the middle finger will indicate the direction of induced current in the conductor as shown below..



It is used to direction of induced EMF and current in generators.

Flat Hand Rule: It is also known as the Right Hand Palm Rule. It is used to determine the direction of the magnetic force on a current-carrying conductor in a magnetic field. With the thumb pointing in the direction of the current and the fingers pointing in the direction of the magnetic field, the force is then indicated by the direction the palm of the hand is facing as shown below.



Dynamically Induced EMF: It is an EMF generated when a conductor moves through a stationary magnetic field. This phenomenon takes place in electric generators and back emf of motors and also in transformers.

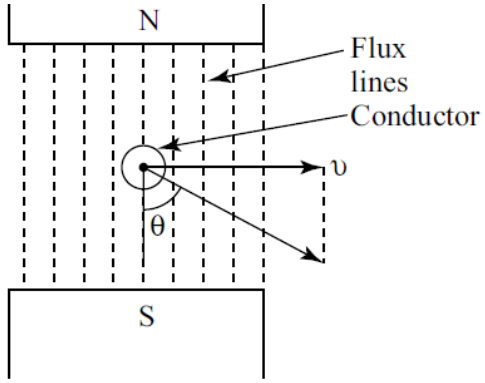


Fig : Conductor moving in a magnetic field

$$e = B\,l\,v\,\sin\theta \quad \textbf{volts}$$

Where

B = Magnetic field strength in wb/m^2

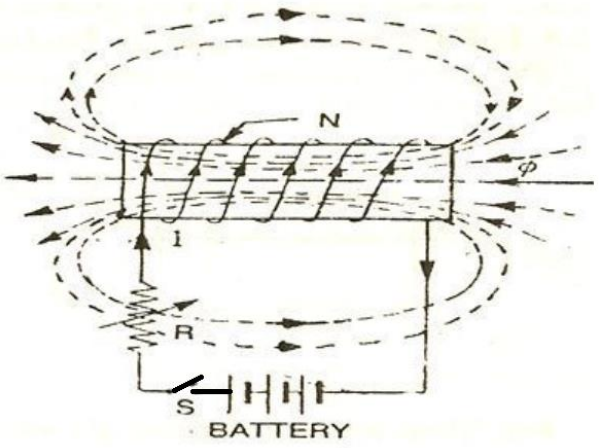
l = Length of the conductor lying in magnetic field

v = Velocity of the conductor in metre/second or m/s

Θ = Angle at which conductor moves in a direction with magnetic field

Statically Induced EMF : The EMF is generated by keeping the coil and the magnetic field system stationary at the same time; and the change of flux linking with the coil is obtained by changing the electric current through it.

a) Self-Induced EMF: It is a type of emf induced (e) in the coil due to the change of flux produced by it linking with its own turns.



N = Number of Turns of the Coil

I = Current flowing through the coil

S = Switch used to change the current through the coil

ϕ = Magnetic Flux in webers (wb)

Fig : Self Induced EMF

$$e \;\alpha\; N\frac{d\emptyset}{dt} \;\alpha\; \frac{dI}{dt}$$

$$e = -N\frac{d\emptyset}{dt} = -L\frac{dI}{dt}$$

where

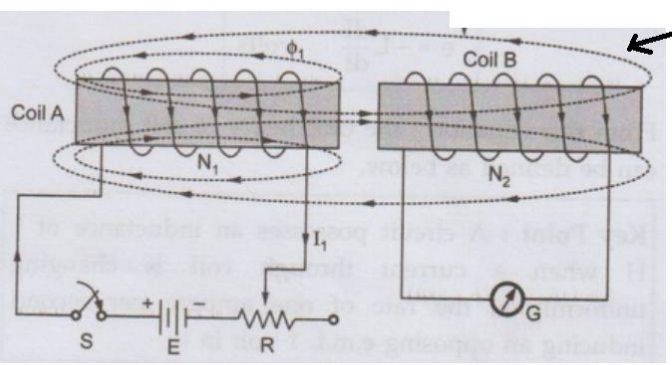
L = Self inductance of the coil = $\frac{N\emptyset}{I}$ Henry or H

The negative sign is due to **Lenz's law**.

Self inductance (L) of the coil: It is defined as the property of the coil by which e.m.f. gets induced in it because of change in current through it.

A coil is said to have inductance of 1 H when 1 A current flows through it produced flux linkages of 1 wb-turn in it.

b) Mutually Induced EMF: It is a type of emf induced in a coil due to the change of flux produced by another neighbouring coil linking with it (ϕ_{12}).



Let

N_1 = Number of turns of coil A,

N_2 = Number of turns of coil B

I_1 = Current flowing through coil A

Φ_1 = Flux produced due to current I, in webers.

ϕ_{12} = Flux linking with coil B due to change current through the coil 1, called mutual flux

Fig : Mutually Induced EMF

$$e = -N_2 \frac{d\emptyset_{12}}{dt} = -M \frac{dI_1}{dt} \quad volts$$

where

$$M = \frac{N_2 \emptyset_{12}}{I_1} \quad Henry\ or\ H$$

The mutual inductance between coils is said to be 1 H, when 1 A current flows through first coil produces flux linkages of 1 wb-turn in neighbouring coil.

Mutual inductance (M) : It is defined as the property of the coil by which e.m.f. gets induced in the second coil because of change in current through first coil.